SAE Aero Capstone SSR

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The SAE Aero Design Competition: *40 years in the making*

Students create an airplane that…

-Is remote controlled
-Pushes participants ingenuity
-Meets dimensional and weight requirements
-Uses the allowed materials
-Meets performance standards



Problem Statement

"The Regular Class objective is to design a multi-motor cargo aircraft that can maximize mission performance associated with payload delivery."

Scoring:

Overall Final Flight Score (OFFS): Average of best three flights, more payload = more points (incentivized incremental gains), payload prediction bonus

Timed Operational Requirements: 100 ft takeoff, 400 ft in-flight straightaway, 360° turn, 400 ft landing, 60 second payload unloading

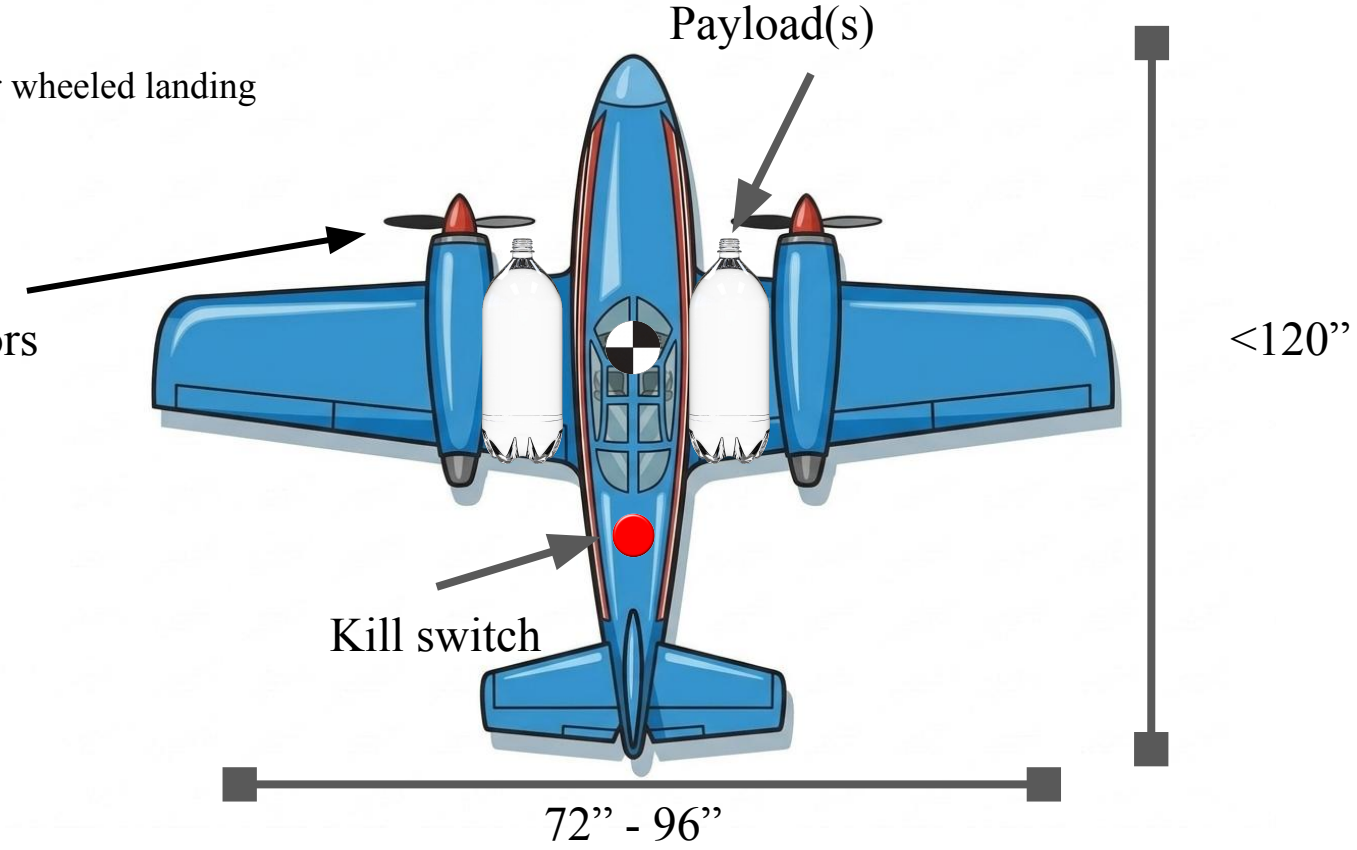
Nonflight Technical Deliverables: Technical Design Review, Flight Demonstration Readiness Review

General Requirements

< 55 lbs (payload included)

No fiber reinforced plastic (duct tape) or rubber bands

Steering required for wheeled landing gear



Design Specifications

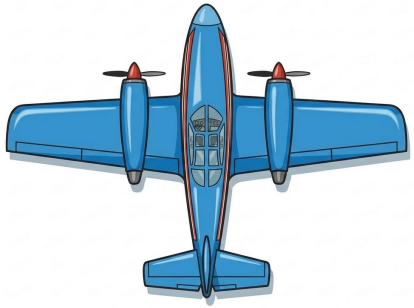
The specified design requirements are in accordance with the SAE 2027 Regular Class Competition rules, the main rules that need to be adhered are in the following tables

Rule #	Parameter	Requirement
2.5	Gross Weight	Gross take-off weight not to exceed 55lbs.
7.1	Dimension Requirement	Minimum wingspan 72 in, maximum wingspan 96 in, wing chord greater than 4 in, maximum body length less than 120 in.
7.2	Material and Equipment Restrictions	No Fiber-Reinforced Plastic (FRP) (duct tape) on any part of aircraft; exceptions include commercially available FRP motor mount, propeller, landing gear, and control linkage components.

Design Specifications

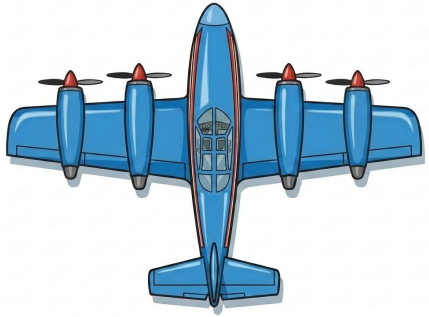
Rule #	Parameter	Requirement
7.3	Propulsion System Requirement	2 or 4 Electric Motors; no restrictions on make or model. Powered by a commercially available 4-cell (14.8V) Lithium-Polymer battery pack; max requirement 2200 mAh; Max propeller diameter 12 in for two motors and 9 for 4 motors.
7.4.0.0	Payload Requirement	Consists of unmodified commercially available 2L cylindrical plastic bottles, carried internally.
7.4.1.2	Payload Requirement	Bottle Cargo has two configs: EMPTY BOTTLE: Weight between 1-4lbs. FILLED BOTTLE: Weight greater than or equal to 4lbs.

Design Consideration: Two Motors vs. Four Motors



2 Motors

- 12” max propeller
- Lower Weight
- Lower drag
- Simpler wiring
- Higher load on each motor
- Easier Maintenance
- 226 in^2 maximum total propellor disk area



4 Motors

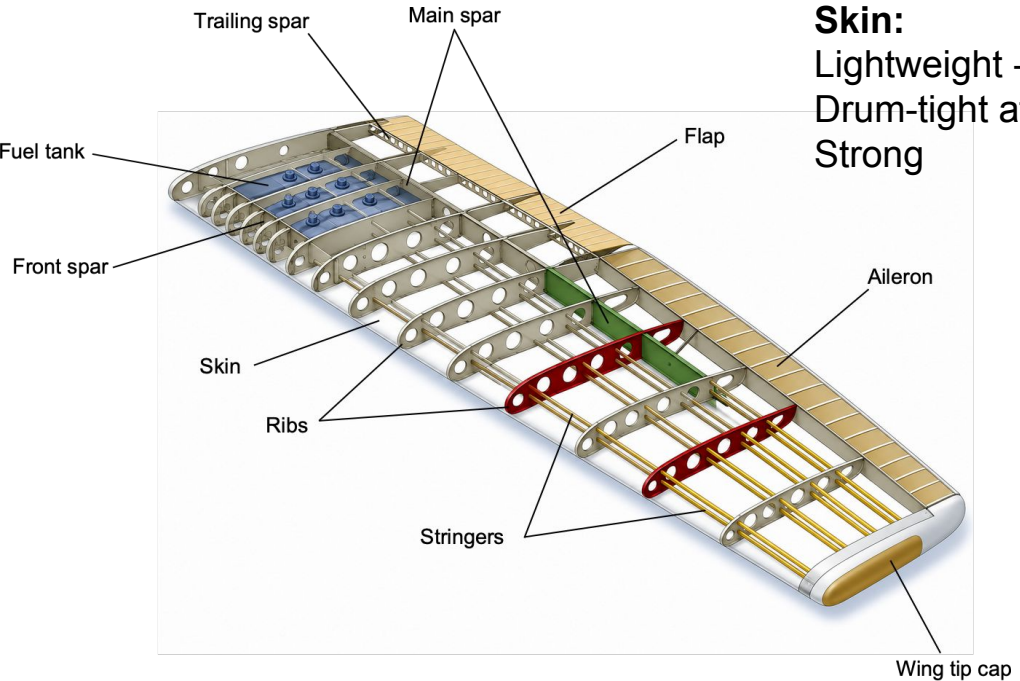
- 9” Max propeller
- More Power Draw from Battery
- More Weight
- More Power/High Takeoff Thrust
- More Maintenance
- Redundancy mitigates failure
- 255 in^2 maximum total propellor disk area

Preliminary Direction:

2 Motors:

- Larger props
- Lower mass
- Simpler integration

Design Considerations: Airframe



Skeleton:
Lightweight - Strong (non-carbon) = **Balsa Wood**
Legacy use
6-12 lbm/ft^3

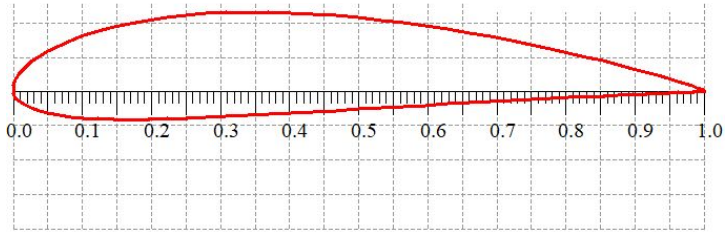
Skin:
Lightweight - Strong (non-duct tape) = **Mylar Film**
Drum-tight after utilizing heat shrink properties
Strong

Design Considerations: Wing Design

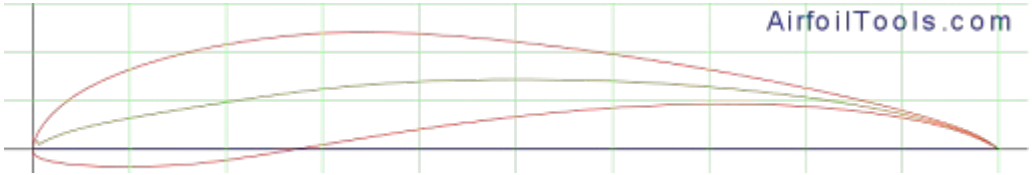
High Lift - Low Drag (Lift Coefficient & Lift Drag Ratio); Buildability

50,000 < Re < 200,000 (RC aircraft)

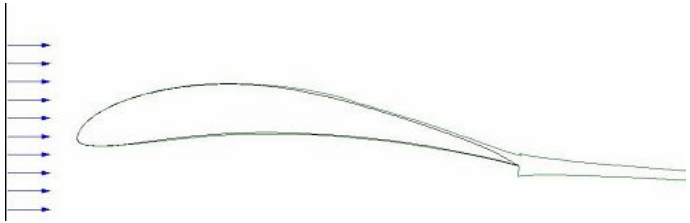
NACA 4415 High Lift to Drag Ratio



Selig s1223: Low Re Lift



Eppler 423: Most lift



Design Consideration: Cargo Bay

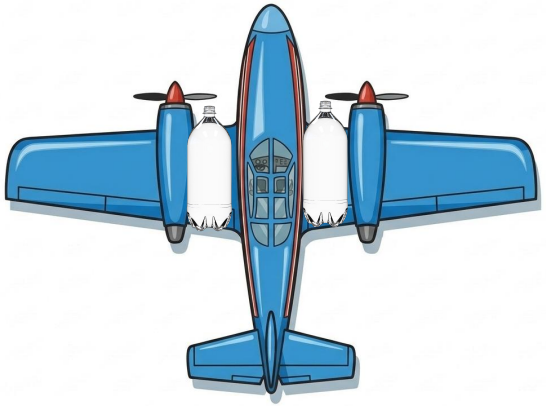
Door & Latch Mechanism



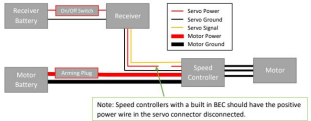
Bomb Bay



Wing Mounted



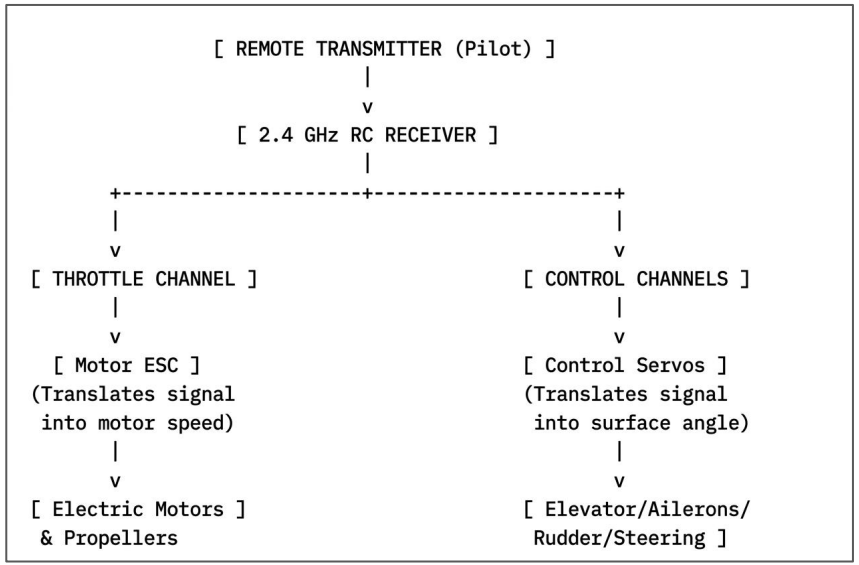
*Tentative due to unknown fuselage conditions

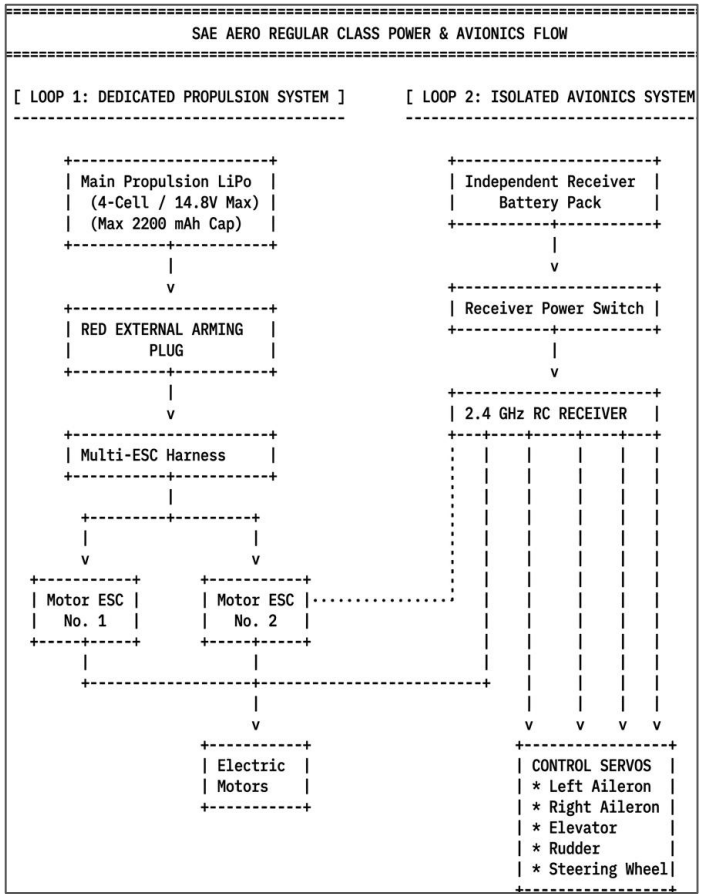


Design Consideration: Power + Control

Under **Rule 2.23 and Rule 7.3**, the propulsion power loop and the flight control loop must remain electrically isolated.

Rule 2.22: BIG RED BUTTON  9” away from propellers

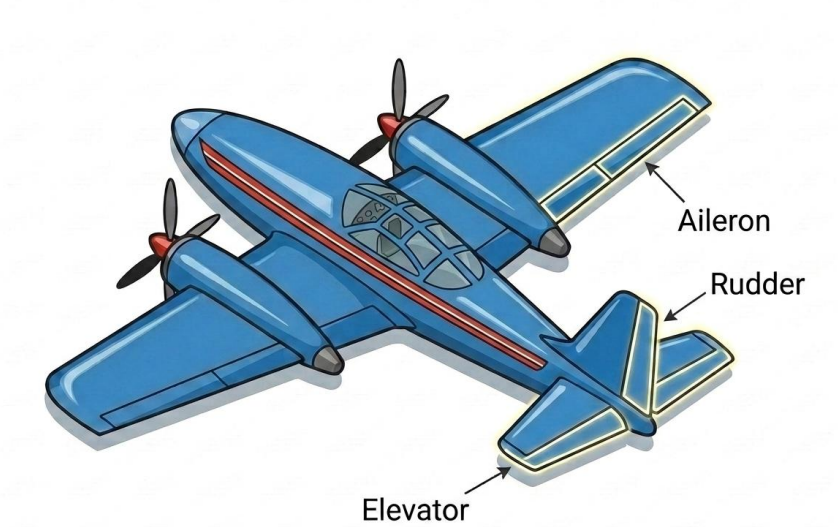




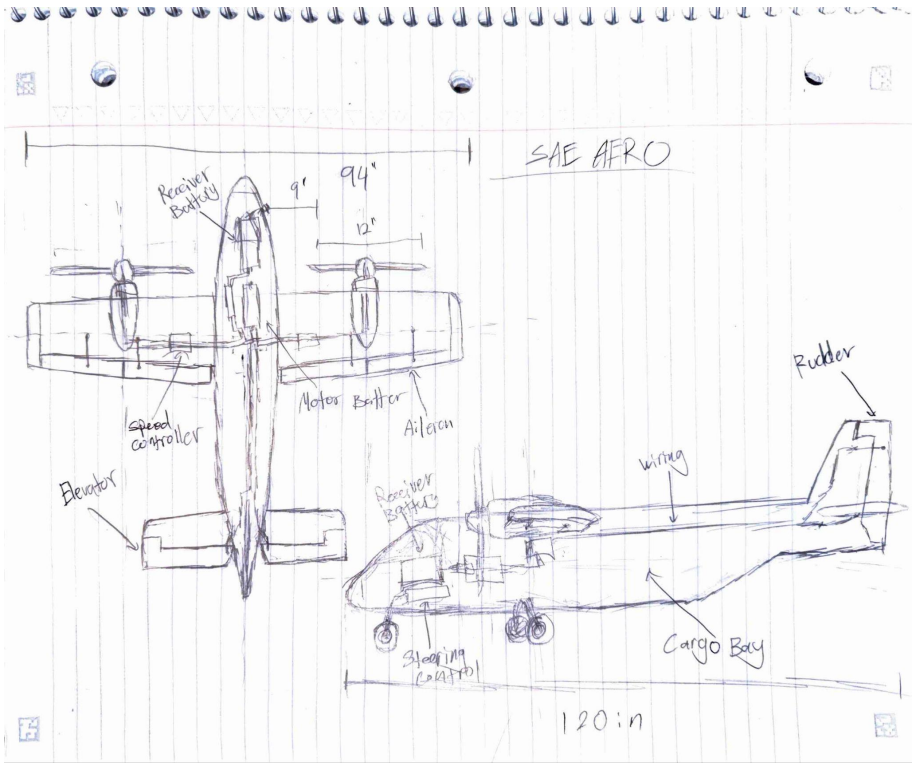
Design Consideration: Actuators & Steering

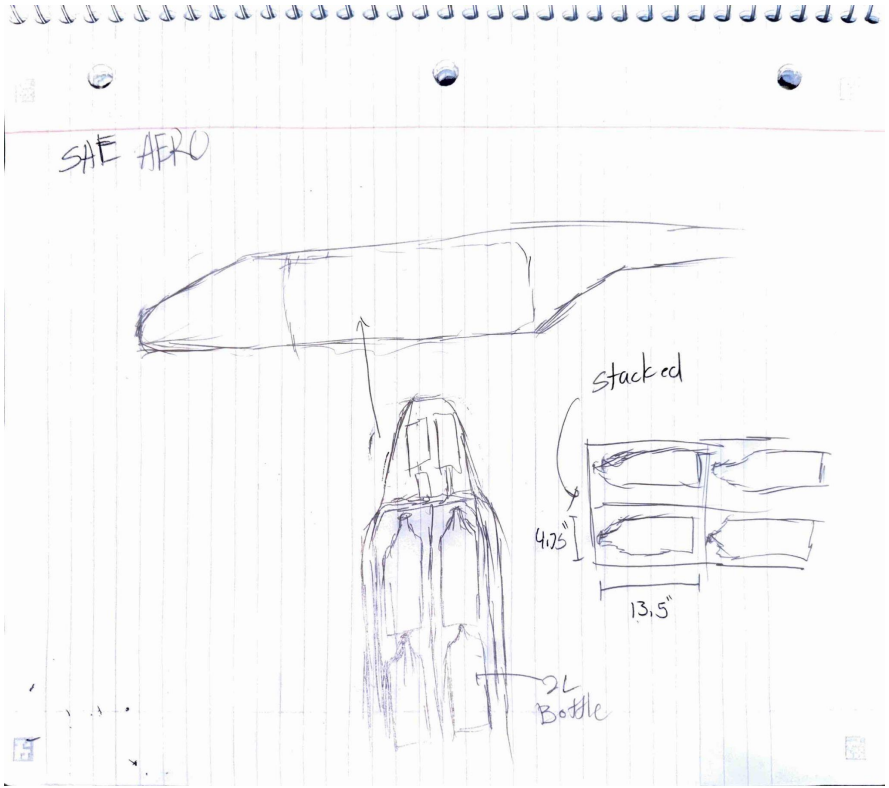
Actuated components: rudder, elevator, aileron, front wheel steering

General Perceived Approach: Servo -> Pushrod -> Control horn



Design Sketches





Timeline

Task List

add links when neccesary

Subsystem	Deliverable	Lead	Support	Start	Due	Status	Links	Days Left	Schedule
Documentation	Finish SRR chapters I–IV	Team	Team	Sep 17, 2026	Sep 21, 2026	Done	SRR		Complete
Documentation	Finish SRR presentation slides	Team	Team	Sep 17, 2026	Sep 23, 2026	Done			Complete
Wing	Compare airfoils	Diego	Team	Sep 22, 2026	Oct 14, 2026	Not Started		16	On Track
Payload	Compare bottle layouts	Lucas	Kasin	Sep 22, 2026	Oct 14, 2026	Not Started		16	On Track
Propulsion	Compare two and four motors	Cooper	Kasin Taran Lucas	Sep 22, 2026	Oct 14, 2026	Not Started		16	On Track
Wing	Estimate wing area	Taran	Diego			Not Started			No date
Controls	Compare radio and receiver	Cooper	Kasin		Oct 14, 2026	Not Started		16	On Track
Power	Draft power distribution diagram	Cooper	Team			Not Started			No date
Structures	Estimate loaded CG	Kasin	Lucas			Not Started			No date
Fuselage	Estimate fuselage size	Kasin	Lucas			Not Started			No date

Thank you :D


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